Andrew R. Marderstein

Research Fellow | Memorial Sloan Kettering Cancer Center

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Twitter: @amarderstein | GitHub: github.com/drewmard

Education

Postdoctoral Fellow, Stanford University — 2021 – 2024

School of Medicine Dean's Fellow in the Department of Pathology.
PI: Stephen Montgomery. Close collaborators: Ana Cvejic (Cambridge/Copenhagen), Anshul Kundaje (Stanford).

PhD - Computational Biology & Medicine (CBM), Tri-Institutional Program — 2017 – 2021

Cornell University / Weill Cornell Medical College | Memorial Sloan Kettering Cancer Center | Rockefeller University
PIs: Andrew Clark (CU) and Olivier Elemento (WCMC).

BS, Cornell University — 2013 - 2017

Major in Biometry & Statistics (Concentration: Statistical Genomics) and a Minor in Biological Sciences.
PI: Philipp Messer. Graduated with honors and distinction in research. Departmental (Major) GPA: 4.00.

Research Experience

Memorial Sloan Kettering Cancer Center — 2024 – Present
Andrew Sabin Family Foundation Fellow, Division of Clinical Genetics (led by Dr. Kenneth Offit)
> Studying high-risk cancer patients, supported by $150,000/year in fellowship funding for research support

Stanford University — 2021 – 2024
School of Medicine Dean's Postdoctoral Fellow, Dr. Stephen Montgomery Lab
> Combining human genetics, single-cell multi-omics, and AI/ML to understand disease mechanisms

Pfizer (Biomarker & Precision Medicine Team), **Third Rock Ventures** (Marea Therapeutics) — 2021 – 2024
External Consultant (while at Stanford)
> Predicting clinical outcomes, disease progression, and new drug targets from genomic datasets

Cornell University and Weill Cornell Medical College — 2017 - 2021
PhD Candidate; Dr. Andrew Clark Lab and Dr. Olivier Elemento Lab
> Thesis: Analyzing modifiable risk factors in disease using human genetic variation (e.g. GxE interactions)

Regeneron Genetics Center — Summer 2016/Summer 2017
Statistical Genetics Intern; Supervised by Dr. Cristopher Van Hout and Dr. Shane McCarthy
> Evaluated the false positive rate and power of statistical methods commonly-used for GWAS

Cornell Department of Biological Statistics and Computational Biology — 2015 – 2017
Undergraduate Honors Researcher; Dr. Philipp Messer Lab
> Developed a Bayesian coalescent framework for studying selective sweeps in population genetic data

Cornell Department of Natural Resources — 2016 – 2017
Statistician – Consultant for Hudson River Fisheries Unit; Supervised by Dr. Pat Sullivan
> Created quantitative models of fish population dynamics in time-series data

Cornell Institute for Resource Information Sciences (IRIS) — 2015 – 2016
Undergraduate Researcher; Supervised by Dr. Magdeline Laba
> Developed New York State 4-H youth education programs; mapped invasive species in the Hudson River

Presented Work

Publications: (^ or * = co-first or co-corresponding authors)

Published/Accepted:

1. **Marderstein, A.R.,** Montgomery, S.B. (2026). Non-coding variant prioritization based on cell type, developmental stage, and evolutionary constraint. *Nature Genetics*. doi: 10.1038/s41588-026-02618-7. (Research Briefing)

2. **Marderstein, A.R.^***, Kundu, S.^, …., Kundaje, A.*, Montgomery, S.B.* Decoding common and rare non-coding variant effects across cellular and developmental contexts. *Nature Genetics*. doi: https://doi.org/10.1038/s41588-026-02619-6

3. Uppal, M., Bhinder, B., **Marderstein, A.R., …,** Salles, G.*, Joffe, E.* (2026). Integrative molecular analysis reveals determinants of clinical outcomes in *TP53*-mutated diffuse large B-cell lymphoma. *Journal of Clinical Oncology*. doi: 10.1200/JCO-25-01928

4. Rubenstein, A.B., Smith, G.R., Zhang, Z., …, **Marderstein**, **A.R.,** Montgomery, S.B., …, Trappe, S.*, Sealfon, S.C.* (2025). Integrated single-cell multiome analysis reveals muscle fiber-type gene regulatory circuitry modulated by endurance exercise. *Genome Research*. doi: 10.1101/gr.280051.124

5. Coorens, T.H.H., et al., the **dGTEx Consortium** (2025). The human and non-human primate developmental GTEx projects. *Nature*. **637**, 557–564 (2025). https://doi.org/10.1038/s41586-024-08244-9

6. **Marderstein, A.R.**, De Zuani, M., Moeller, R., Bezney, J., Padhi, E., Wong, S., Coorens, T.H.H., Xie. Y., Xue, H., Montgomery, S.B.*, Cvejic, A* (2024). Single-cell multi-omics map of human fetal blood in Down's Syndrome. *Nature* 634, 104-112. https://doi.org/10.1038/s41586-024-07946-4
   - > *Nature* News & Views: "Childhood leukaemia in Down's syndrome primed by blood-cell bias"
   - > *Nature* Podcast: "Children with Down's Syndrome are more likely to get leukaemia: stem cells hint at why"

7. **IGVF Consortium** (2024). Deciphering the impact of genomic variation on function. *Nature* 633, 47–57. doi: https://doi.org/10.1038/s41586-024-07510-0

8. Lima, S., Rupert, A., Putzel, G., **Marderstein, A.R.**, …, Scherl, E., Longman, R (2024). The gut microbiome regulates the clinical efficacy of sulfasalazine therapy for IBD-associated spondyloarthritis. *Cell Reports Medicine*. doi: https://doi.org/10.1016/j.xcrm.2024.101431.

9. Shah, Y.^, Verma, A.^, **Marderstein, A.R.**, …, Elemento, O (2021). Pan-cancer analysis reveals molecular patterns associated with age. *Cell Reports* 37, 110100. doi: https://doi.org/10.1016/j.celrep.2021.110100.

10. **Marderstein, A.R.**, Kulm, S., Peng, C., Tamimi, R.M., Clark, A.G.*, Elemento, O.* (2021). A polygenic score-based approach for identification of gene-drug interactions stratifying breast cancer risk. *American Journal of Human Genetics* 108 (9), 1752-1764. doi: 10.1016/j.ajhg.2021.07.008.
    - > *GenomeWeb*: "Breast Cancer Genetic Risk Modified by Corticosteroid Use"

11. **Marderstein, A.R.**, Davenport, E.R., Van Hout, C.V., Kulm, S., Elemento, O.*, Clark, A.G.* (2021). Leveraging phenotypic variability to identify genetic interactions in human phenotypes. *American Journal of Human Genetics* 108 (1), 1-19. https://doi.org/10.1016/j.ajhg.2020.11.016
    - > *Cornell Chronicle*: "Cross-campus team probes gene-environment interactions"

12. **Marderstein, A.R.**, Uppal, M., Verma, A., Bhinder, B., Tayyebi, Z., Mezey, J., Clark, A.G.*, Elemento, O.* (2020). Demographic and genetic factors influence the abundance of infiltrating immune cells in human tissues. *Nature Communications* 11, 2213. https://doi.org/10.1038/s41467-020-16097-9

13. Castellanos, J.G., …, **Marderstein, A.R.**, … , Scherl, E.J., Longman, R.S. (2018). Microbiota-induced TNF-like ligand 1A drives Group 3 innate lymphoid cell-mediated barrier protection and intestinal T cell activation during colitis. *Immunity*. 49(6):1077-1089.e5. doi: 10.1016/j.immuni.2018.10.014

14. **Marderstein, A.R.** (2017). Approximate Bayesian Computation for Studying Selective Sweep Signatures in Local Coalescence Trees. Honors Thesis, Cornell University. Supervisor: Philipp Messer.

Under Review/In Revision/Submitted/In Prep:

1. **Marderstein, A.R.^***, Hunt, G.^, Padhi, E.M.^, …., Cvejic, A.* Trisomy 18 and 21 disrupt lineage-specific regulatory networks in foetal haematopoiesis. (In revision)
2. Niu, Y., Fansler, M., Chen, W, …, **Marderstein, A.R.**,…, Cvejic, A. The epigenetic and transcriptomic blueprint of human foetal liver haematopoiesis across development. (In revision at *Nature*)

3. Amsalem, J., Ostrovnaya, I., **Marderstein, A.R.**, …, Offit, K.*, Joseph, V.* Integrated Clinicogenomic Risk Modeling for Metachronous Second Primary Cancers. (Under review at *Nature Communications*). Preprint on *medRxiv*. doi: https://doi.org/10.64898/2026.06.12.26355388
4. **The ENCODE Project Consortium**. The Encyclopedia of DNA Elements. (Under review at *Nature*)
5. Perea-Chamblee, T.^, Sun. H.^, …, **Marderstein, A.R.**, …, Gusev, A.*, Carrot-Zhang, J.*. Multi-ancestral real-world data reveals germline determinants of somatic drivers and cancer progression. (In revision)
6. Kulm, S., **Marderstein, A.R.**, Mezey, J.*, Elemento, O.* A systematic framework for assessing the clinical impact of polygenic risk scores. Preprint on *medRxiv*. doi: https://doi.org/10.1101/2020.04.06.20055574 (In revision)
7. Krishna, C., Walker, R.L., **Marderstein, A.R.**, …, Xavier, R.J. Immunogenetic control of gene expression and cell state. (In revision)
8. Kong, Y., Harrington, D., **Marderstein, A.R.**, Alonso, L.C. Low frequency T2D-associated polymorphisms at *CCND2* influence *CCND2* mRNA abundance, but not proliferation, in human pancreatic islets. (In revision)
9. Liu, J., …, **Marderstein, A.R.**, …, Natarajan, P.*, Offit, K.*, Bolton, K.L.* Characterization of the Cancer Spectrum across 92,384 Germline Cancer Predisposition Carriers. (In prep, July '26)
10. Padhi, E., …, **Marderstein, A.R**, …, Lennon, N.*, Montgomery, S.B.* Multi-omics in 10,000 diverse individuals expands mapping genetic mechanisms in health and disease. (In prep, August '26)
11. **Marderstein, A.R.***, Banaszak, L.*, Shin, N.Y.*, Offit, K., Cantor, A.B. Clinical and mechanistic dissection of non-coding variants in the *ANKRD26* 5'UTR using deep learning. (ASHG Abstract '26; In prep)
12. Keshavjee, S.*, **Marderstein, A.R.***, …, Carrot-Zhang, J., Stadler, Z., Abass, M.A. Polygenic Risk Score Contribution to Colorectal Cancer Risk in Lynch Syndrome. (CGA-IGC Abstract '26; In prep)

Platform Presentations:

Conference:

[Presenter]

2026	Emerging Scholars in Genome Sciences Symposium, University of Virginia, VA (*Upcoming Talk*)
2026	Hereditary Cancer Genetics: Incorporation into Clinical Care Symposium, MSKCC, NY (*Upcoming Talk*)
2026	Robert Steel Symposium in Developmental Oncology, Memorial Sloan Kettering, NY
2026	University of Pennsylvania Emerging Genomic Scientists Symposium, Philadelphia, PA
2025	Alzheimer's Disease Sequencing Project - Functional Genomics Consortium Meeting (co-author, presented by Soumya Kundu at Gladstone Institutes, San Francisco, CA)
2025	Developmental GTEx Project Spring Meeting at the Broad Institute, Cambridge, MA
2024	American Society of Human Genetics conference in Denver, CO (Platform title: "Genetics of Human Brain: Regulation, Disease Risk, and Assortative Mating")
2023	American Society of Human Genetics conference in Washington D.C. (Platform title: "Causal non-coding variants and the genes they impact")
2023	IGVF Annual Meeting in St. Louis, MO
2022	American Society of Human Genetics conference in Los Angeles, CA (Platform title: "Genetic Impacts on the Epigenome and Beyond") > **Semifinalist: 2022 Charles J. Epstein Trainee Awards for Excellence in Human Genetics Research**
2022	dGTEx Annual Meeting in Bethesda, MD (on behalf of the Data Analysis & Assays WG)
2022	IGVF Annual Meeting in Bethesda, MD (on behalf of the Immune Disease FG)
2020	eKeystone Virtual Symposia: Beyond a Million Genomes (*Invited Talk*) (Virtual)
2020	Beyond a Million Genomes: From Discovery to Precision Health, Keystone Symposia, Breckenridge, CO.
2019	American Society of Human Genetics conference in Houston, TX. (Platform title: "Mechanisms of Immune Cell Phenotypes and Clonal Hematopoiesis") > **Semifinalist: 2019 Charles J. Epstein Trainee Awards for Excellence in Human Genetics Research**
2019	New York Academy of Sciences: Translating Genetics into Medicine Conference in New York, NY.
2018	Population, Evolutionary, and Quantitative Genetics conference in Madison, WI.

[Senior author]

2022	American Society of Human Genetics conference in Los Angeles, CA (led by Lin Poyraz) (Platform title: "The current environment for gene-environment interactions")
2021	Society for Molecular Biology & Evolution (led by Lin Poyraz) (Platform title: "Genotype-environment interactions in the genomics era")

Internal:

2026	Lymphoid Malignancy Translational Research Meeting Series, MSKCC, New York, NY (*Upcoming Talk*)
2025	New York Genome Center, New York, NY (*Invited Talk*)
2025	Biomedical Data Science Innovation Lab Seminar Series, University of Virginia, VA (Virtual *Invited Talk*)
2025	Cornell University, Ithaca, NY
2024	University of Chicago, Chicago, IL

2024	Biotech Research Innovation Centre (BRIC), University of Copenhagen, Denmark (*Invited Talk*)
2024	Case Western Reserve University, Cleveland, OH
2024	Inst. of Stem Cell Biology & Regenerative Medicine at Stanford University, Stanford, CA
2024	Memorial Sloan Kettering Cancer Center, New York, NY
2024	Columbia University, New York, NY
2023	Yale University, New Haven, CT (*Invited Talk*)
2023	Center for Human Genetics & Genomics, NYU, New York, NY
2023	Division of Precision Medicine, Department of Medicine, NYU, New York, NY
2023	IGVF: Non-coding Variants FG (Virtual), IGVF Immune Disease FG (Virtual)
2023	Pediatric Computational Biology Group at Stanford University, Stanford, CA
2022	IGVF: Computation, Analysis, Modeling, Prediction WG (Virtual), Phenotypic Impact FG (Virtual), Immune Disease FG (Virtual)
2021	Tri-I Computational Biology & Medicine Seminar at Weill Cornell Medicine (Virtual)
2021	Student Talk at Tri-I Computational Biology & Medicine Program Recruitment (Virtual)
2021	Department of Computational Medicine, UCLA (*Invited Talk*) (Virtual)
2020	Analytical & Translational Genetics Unit, Broad Institute (*Invited Talk*) (Virtual)
2020	Student Talk at Tri-I Computational Biology & Medicine Retreat (Virtual)
2020	Tri-I Computational Biology & Medicine Seminar at Weill Cornell Medicine, New York, NY
2020	Computational Biology Student Seminar at Cornell University, Ithaca, NY
2019	Basic Research Working Group meeting at Weill Cornell Medicine, New York, NY
2019	Panelist, Precision Medicine Symposium at Weill Cornell Medicine, New York, NY (Panel discussion title: "Facilitating Cross Campus Collaboration for Trainees")
2019	Microbiome Meeting at Cornell University, Ithaca, NY
2019	Invited alumni panelist, Cornell Accepted Students Reception in Rye, NY
2019	Computational Biology Student Seminar at Cornell University, Ithaca, NY
2017	Regeneron Genetics Center in Tarrytown, NY
2016	Intern B&B, Regeneron Pharmaceuticals, Inc in Tarrytown, NY
	> **Intern project was selected to be part of a company-wide telecast and presentation**

Poster Presentations:

Conference:

2026	ASHG Annual Meeting in Montreal, Canada (*Upcoming*)
2024	IGVF Annual Meeting in Seattle, WA
2023	IGVF Annual Meeting in St. Louis, MO
2023	Biology of Genomes conference in Cold Spring Harbor Laboratory, NY
2021	American Society of Human Genetics conference (held virtually due to the COVID-19 pandemic)
2020	American Society of Human Genetics conference (held virtually due to the COVID-19 pandemic)
	> **Awarded "Reviewer's Choice" for top 10% of all submitted abstracts**.
2020	Beyond a Million Genomes: From Discovery to Precision Health, Keystone Symposia in Breckenridge, CO
2019	New York Academy of Sciences: Translating Genetics into Medicine Conference in New York, NY
	> **Awarded the "F1000 Outstanding Presentation Prize" for best poster.**
2018	American Society of Human Genetics conference in San Diego, CA
2018	Population, Evolutionary, and Quantitative Genetics conference in Madison, WI

Local:

2020	Tri-I Symposium in Computational Biology & Medicine at Memorial Sloan Kettering, New York, NY
2019	Annual Vincent du Vigneaud Student Research Symposium at Weill Cornell Medicine, New York, NY
2017	Honors Symposium, Cornell University in Ithaca, NY
2016	Cornell Undergraduate Research Board Fall Forum in Ithaca, NY

Leadership and Service

Program Committee Member, **American Society of Human Genetics** 2025-2027

Speaker, **Skype a Scientist** 2017 – Present

- Chat about fascinating genetics research with 7th to 12th grade classrooms, with an emphasis on captivating students' interests and clearly explaining complex topics to students that may have limited scientific backgrounds. I've engaged over 1,000 students total around the country through this program.

Fundraiser, **Crohn's & Colitis Foundation of America (CCFA)** 2011 – Present

- Fundraising to promote awareness and find a cure. Half-marathon Team Challenge finisher.

Co-Chair of the Career Development Committee, **American Society of Human Genetics** 2020 – 2024

- Organize, moderate, and lead career fairs, panels, workshops, and resources for early-career members, such as:
- Host of the "Trainee Paper Spotlight" podcast, bringing stories from up-and-coming scientists into the spotlight
- Lead organizer of the "Entrepreneurship in Human Genetics" panel at the 2020 meeting
- Co-organizer and panelist for the "Tips on How to Land a Postdoctoral Position" at the 2021 meeting
- Founded the "Trainee Social Reception" at the 2022 meeting; now a yearly event (>400 attendees)
- Co-organizer of "The Confidence Factor: Thriving in Your Career" at the 2023 meeting
- Co-organized a bimonthly "Work-Life Balance" webinar series on rotating themes from 2021-2023
- Co-organizer and moderator of "How To Set Yourself Apart" panel at the 2024 meeting

2023 IGVF Annual Meeting Planning Committee, Impact of Genomic Variation on Function (**IGVF)** 2023

- In addition to helping organize the annual meeting for 200+ attendees, I moderated a session and led a break-out discussion during the meeting.

IGVF Immune Trainee Focus Group Co-Lead, **IGVF** 2023

IGVF All-Consortium Monthly Meeting Organizing Committee, **IGVF** 2023

2023 Annual Meeting Abstract Reviewer, **American Society of Human Genetics** 2023

- Section: Complex Traits and Polygenic Disorders III: Infectious disease and immunological disorders

Focus Group for evaluating a professional Certification Program, **American Society of Human Genetics** 2023

2022 Meeting Planning Advisory Group, **American Society of Human Genetics** 2022

- Provide counsel to the staff in the face of opportunities, challenges, and costs of a return in a new environment
- E.g. how to communicate vaccination requirements, or how to deliver hybrid (remote/in-person) content

Search Committee for Diversity, Inclusion and Belonging Position, **Stanford Genetics Department** 2022

- Postdoctoral Representative of the committee helping build a diverse and inclusive community at Stanford

Judge, Amgen Scholars - **Stanford Summer Research Program (SSRP) Symposium** 2022

Lead Organizer, **Montgomery Lab Retreat** in Lake Tahoe, CA 2022

Guest Lecturer, **BIOS 2000: "Foundations in Experimental Biology"**, Stanford University 2021

President, **Tri-Institutional Biotech Club** 2019 – 2021

- The largest club on campus (includes students and postdocs across Cornell, Sloan Kettering, and Rockefeller)
- Led organization & moderated ~15 major Tri-I-wide events with industry professionals (100 attendees each) [e.g. "Human Genetics in Drug Development", "Biomedical Innovation in a Non-Profit Organization"]
- Envisioned, recruited, founded, and led the Symposia Team of 12 new members, in order to develop the first Biotech symposium across the Tri-I (to be held in Fall 2021).

Student Leadership, **Tri-Institutional CBM PhD Program** 2018 – 2021

- Student Representative, CBM Curriculum Committee (2018); Organizer, Research-in-Progress Seminars (2018-2019); Organizer, CBM Annual Retreat (2019); Head of Social Media (2019 – Present); Summer Internship Admissions Committee (2020)

DNA Day Essay Contest – Round 2 Judge, **American Society of Human Genetics** 2020

Tutor, **Grade Expectations** 2018 – 2019

- Teach introductory statistics to high school and college students in the NYC area

Abstract Committee Member, dVRS Research Symposium 2019**, Weill Cornell Medicine** 2018 – 2019

Pen Pal, **Letters to a Pre-Scientist** 2018 – 2019
> Exchanged monthly letters with 8th grade students interested in biomedical research

Moderator, **American Society of Human Genetics Annual Meeting** (ASHG), San Diego, CA 2018
> Platform Session Title: "Scalable Tools to Enable Collaboration and Reproducible Analyses"

Grader, **BTRY 4840/6840: "Computational Genetics & Genomics"**, Cornell University 2017

Research Judge, **Cornell Undergraduate Research Board Fall Forum** 2017

Athlete & Two-Year Captain, **Cornell University Alpine Ski Team** 2013 – 2017

Biometry & Statistics Peer Advisor, **Cornell Dept Biological Statistics & Computational Biology** 2016 – 2017

Cornell Campus Ambassador, **Regeneron Pharmaceuticals, Inc.** 2016 – 2017

Chair of the New Student Orientation Team, **Cornell Hillel** 2014 – 2016

Awards

Tow Center for Developmental Oncology (TCDO) Career Development Award,
Memorial Sloan Kettering ($450,000 / 3 years) 2025-2028

Andrew Sabin Family Foundation Fellow, Memorial Sloan Kettering ($150,000 / 1 year) 2024-2025

Charles J. Epstein Trainee Award for Excellence in Human Genetics Research (Semi-Finalist),
American Society of Human Genetics Annual Meeting 2022 ($990) 2022

Stanford School of Medicine - Dean's Postdoctoral Fellowship ($32,784) 2022

"Reviewer's Choice" award – scoring in the top 10% of all abstract submissions,
American Society of Human Genetics Annual Meeting 2020 2020

NHGRI Scholarship from Keystone Symposia, Beyond a Million Genomes:
From Discovery to Precision Health conference ($1200) 2020

Charles J. Epstein Trainee Award for Excellence in Human Genetics Research (Semi-Finalist),
American Society of Human Genetics Annual Meeting 2019 ($960) 2019

"F1000 Outstanding Presentation Prize" for best poster, New York Academy of Sciences:
Translating Genetics into Medicine Conference ($1,000) 2019

Finalist, Cornell Health Hackathon 2019

NIH Institutional National Research Service Award (T32), NIGMS - T32GM083937 ($84,000) 2017-2019

Genomic Scholars Program, Cornell Center for Vertebrate Genomics ($5,000) 2018

Conference Travel Grant, Cornell University – American Society of Human Genetics conference ($440) 2018

Conference Travel Grant, Cornell University – Population, Evolutionary, and
Quantitative Genetics Conference ($360) 2018

Robert S. Elster Memorial Scholarship – for "exhibiting leadership and dedication toward
providing outstanding community service to his local community" ($1,500) 2017

Dean's List (7x) 2013-2017

Intern B&B Speaker, Regeneron Pharmaceuticals, Inc. –
one of 6 interns nominated to present summer research in a company-wide telecasted presentation 2016

Special Recognition for Outstanding Geospatial Sciences Exhibit, 4-H Club, New York State Fair. 2012

Completed Upper-Level Graded Coursework (selected)

Stanford University: Genetics, Ethics, and Society (a broad, immersive course led by graduate students)

Weill Cornell Medical College: Computational Systems Biology; Clinical and Research Genomics; Genomic Innovation

Cornell University: Computational Genomics; Quantitative Genetics; Human Genomics; Population Genetics; Precision and Genomic Medicine; Bioinformatics Programming; Bayesian Machine Learning; Data Mining and Machine Learning; Probability Models & Inference; Statistical Computing; Linear Models with Matrices; Theory of Statistics; Biological Statistics I & II; Data Structures and Object-Oriented Programming; Physics I & II

Consortium Membership

dGTEx (Developmental Genotype-Tissue Expression project – Assays Tech & Analysis Working Group) 2022 – 2025

IGVF (Impact of Genomic Variation on Function – Immune Focus Group, Non-coding FG, Phenotypic Impact FG) 2022 – 2025

GREGoR (Genomics Research to Elucidate the Genetics of Rare Diseases) 2022, 2024 – 2025

ENCODE4 (Variant to Function WG) 2021 – 2023

Peer Review

Cell Genomics
Nature Communications
GENETICS
Genome Research
Human Genetics and Genomics Advances
Journal of Crohn's and Colitis
Scientific Reports
BMC Genomics
Frontiers in Genetics

Mentorship

Years	Name	Type	Institution	Education Program
2025-	Johnathan Amsalem	Staff	MSKCC	-
2024-25	Louis Liu	PhD	MSKCC	PBSB (Weill Cornell)
2024-25	Sohaib Hassan	PhD	Stanford University	Biomedical Informatics
2021-23	Pelin (Lin) Poyraz	Undergrad	Cornell University	Biological Sciences, Computational Biology
2022-23	Jon Bezney	PhD rotation	Stanford University	Genetics
2023	Omar Johnson	Visiting PhD	UTMB/Stanford	MD/PhD Combined Degree Program
2022	Gyu Kim	PhD rotation	Stanford University	Biochemistry
2021-22	Jiayu Liang	Masters	Cornell University	Applied Statistics

Professional Organization Membership

American Society of Human Genetics	2018 – Present
Genetics Society of America	2018 – 2019
New York Academy of Sciences	2017 – 2021

References

Stephen Montgomery – Stanford University – smontgom@stanford.edu (Postdoc PI)

Kenneth Offit – Memorial Sloan Kettering Cancer Center – offitk@mskcc.org (Primary MSK Fellow Mentor)

Andy Clark – Cornell University – ac347@cornell.edu (PhD Co-PI)

Olivier Elemento – Weill Cornell Medicine – ole2001@med.cornell.edu (PhD Co-PI)

Ana Cvejic – University of Copenhagen – ana.cvejic@bric.ku.dk (Postdoc External Collaborator)

Philipp Messer – Cornell University – pm544@cornell.edu (Undergrad PI)

Other Interests

Lifetime skier (previous captain of Cornell's alpine ski team); Avid runner (5 marathons; 8 half-marathons; NYC Marathon 3-time finisher, '19 volunteer); Passionate New York sports team fan; Intramural softball; Hiking; Designed a baseball simulation (based on the MLB Showdown card game) as a side project; Entrepreneurship (MSK Entrepreneurship Initiative cohort in '25)